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To: Dr. Sahar Abdelnabi

Principal Investigator, COMPASS Research Group

ELLIS Institute Tübingen & Max Planck Institute for Intelligent Systems

Tübingen, Baden-Württemberg, Germany

Re: Application for Postdoctoral Researcher in AI Safety, Alignment, and Cooperative Agent Systems (COMPASS)

Dear Dr. Abdelnabi,

I am writing to enthusiastically apply for the Postdoctoral Researcher position within your research group, COMPASS (COoperative Machine intelligence for People-Aligned Safe Systems), at the ELLIS Institute Tübingen and the Max Planck Institute for Intelligent Systems. As a final-year PhD candidate in Computer Science (thesis submission imminent), my core research focuses on a problem central to COMPASS’s mission: **how to construct oversight signals that remain informative, ungameable, and robust when capable, self-evolving foundation models and multi-agent teams explicitly optimize against them.**

De-Risking Oversight Signals: Anti-Reward-Hacking & Verifiable RL

In recent work (*Gah et al., under submission to SCAIR 2026*), I formulated an annotation-free process reward utilizing cross-view physical consistency between frozen perception backbones. Rather than immediately training against the reward, I designed a **two-arm validate-before-train protocol** to test whether the supervision discriminated correct reasoning and whether it admitted exploitable shortcuts. This methodology surfaced a concrete *specification gaming* failure mode: predicted regions could exploit recall-heavy metrics by ballooning in spatial area without genuine precision. By redesigning the reward around count-based precision and weighted $F_{0.5}$ metrics ($\rho > 0.72 \rightarrow 0.334$), we secured an ungameable signal that successfully optimized a **Qwen2.5-VL-7B** policy via online **Group Relative Policy Optimization (GRPO)** in VERL. At ELLIS, I am eager to scale this counterfactual stress-testing to multi-agent coding and reasoning systems.

From Single-Agent Rewards to Multi-Agent Anti-Collusion & Deception Detection

Naive consensus mechanisms fail in multi-agent environments because colluding agents can coordinate to satisfy surface-level oversight heuristics. My research proposes to detect deception and collusion using **independent counterfactual checks, metamorphic mutation testing, and invariant sandboxing** rather than superficial agent agreement. By treating agent communication through information-theoretic bottlenecks and causal probes, we can ensure multi-agent coding pipelines (e.g., Code-Agent + Test-Agent + Critic-Agent) remain faithful to human intent under joint reinforcement learning.

Calibrated Semantic Arbitration in Multi-Principal Environments

My doctoral thesis, *Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)*, submitted to *IEEE TPAMI*, develops formal probabilistic calibration (Platt scaling, Bernoulli uncertainty modeling) to dynamically coordinate heterogeneous foundation models under distribution shift. This provides the mathematical foundation for **multi-principal contextual integrity**: arbitrating trust, resolving conflicting claims, and bounding information leakage when autonomous agents represent competing human principals with partially conflicting objectives.

Technical Readiness and Commitment to the ELLIS Ecosystem

I bring hands-on expertise in distributed RL frameworks (**VERL**, **GRPO**, **vLLM**), production agentic orchestration (**LangGraph**, **DeepEval**, **Opik**), and SFT/DPO pipelines. I am excited to leverage the ELLIS Institute's world-class compute budget to pursue ambitious, high-risk/high-reward safety research, publish openly at top venues (NeurIPS, ICML, ICLR), and collaborate deeply across Tübingen's research community.

Thank you for your time and consideration. I look forward to the opportunity to discuss my research vision with you.

Sincerely,

Silas Kwabla Gah